

Predicting Loan Approval

Introduction

The banking and financial sector plays an important role in providing loans to individuals and businesses. Before approving a loan, banks analyze several factors such as applicant income, credit score, loan amount, employment status, and loan term. Manual loan approval analysis can be time-consuming and may sometimes lead to incorrect decisions.

Machine Learning helps financial institutions automate the loan approval process by analyzing historical applicant data and predicting whether a loan should be approved or rejected. These systems improve decision-making, reduce risk, and save time.

In this project, the Loan Prediction Dataset was used to build machine learning models for predicting loan approval status. The dataset contains important information such as age, income, credit score, loan amount, loan term, employment status, and loan approval status.

Different preprocessing techniques such as handling missing values, encoding categorical variables, and feature scaling were applied before training the models. Classification algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) were implemented and compared to evaluate prediction accuracy and model performance.

Figure 1: Dataset Preview

df.head()

...	Age	Income	Credit_Score	Loan_Amount	Loan_Term	Employment_Status	Loan_Approved
0	56	81788	334	15022	48	Employed	0
1	69	102879	781	21013	24	Self-Employed	1
2	46	58827	779	39687	60	Self-Employed	0
3	32	127188	364	16886	24	Unemployed	0
4	60	25655	307	26256	36	Unemployed	0

Problem Statement and Methodology

The main objective of the “Predicting Loan Approval Using Machine Learning” project is to develop a machine learning model that can accurately predict whether a loan application should be approved or rejected based on applicant details.

In this project, Loan_Aproved is considered as the target variable, while features such as age, income, credit score, loan amount, loan term, and employment status are used as input variables.

The dataset contains both numerical and categorical data. Therefore, several preprocessing techniques were applied to improve model performance and prediction accuracy.

The methodology followed in this project includes:

- Loading the Loan Prediction Dataset
- Handling missing values
- Encoding categorical variables
- Feature scaling using StandardScaler
- Splitting the dataset into training and testing data
- Training machine learning classification models
- Evaluating model performance using classification metrics

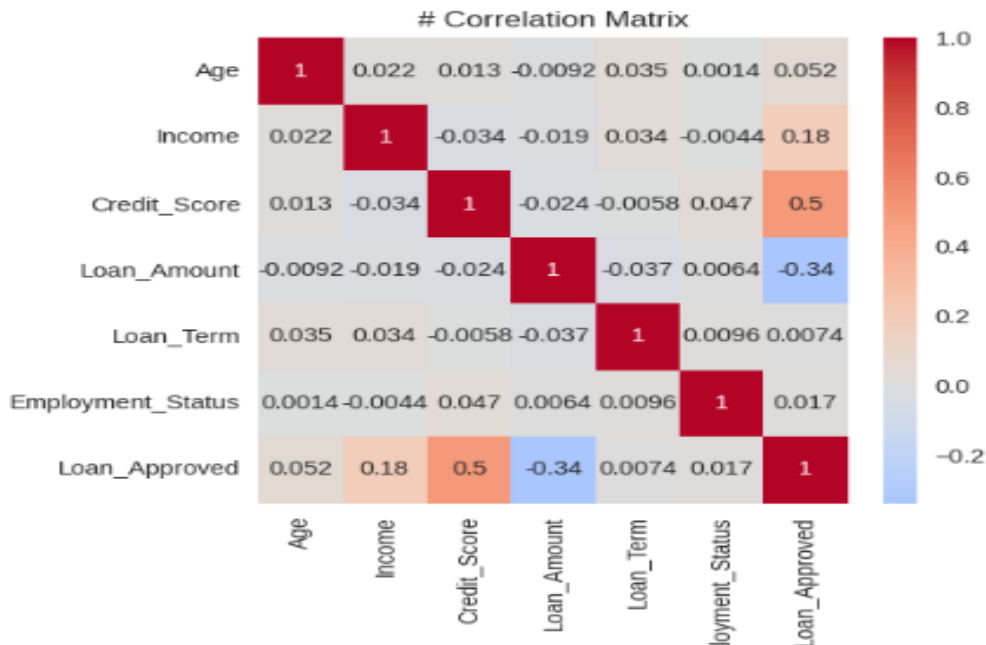
The machine learning models implemented in this project are:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

The project aims to reduce manual effort in loan approval analysis and improve prediction accuracy using machine learning techniques.

Figure 2: Correlation Heatmap

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Results and Discussion

Different machine learning classification models were trained and tested using the Loan Prediction Dataset. The dataset was divided into training and testing data to evaluate model performance.

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

Several preprocessing techniques significantly improved model performance and prediction accuracy. Handling missing values reduced inconsistencies in the dataset, while feature scaling improved model efficiency. Encoding categorical variables also improved training performance.

After comparing the implemented models, Decision Tree and Random Forest achieved the highest prediction accuracy because they handled complex relationships between applicant features more effectively.

Table 1: Model Performance Comparison

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# MODEL PERFORMANCE COMPARISON						
	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
	Logistic Regression	0.8961	0.8768	0.9217	0.8987	0.9698
	Decision Tree	0.9985	1.0000	0.9970	0.9985	0.9985
	Random Forest	0.9985	1.0000	0.9970	0.9985	1.0000
	SVM	0.9729	0.9538	0.9940	0.9735	0.9987

Table 1 compares the performance of different machine learning classification models used for loan approval prediction. Random Forest and Decision Tree achieved the highest prediction accuracy among all the implemented models.

Accuracy Improvements and Insights

Several techniques were applied to improve model accuracy:

- Missing values were handled carefully to avoid data inconsistency.
- Feature scaling improved model efficiency.
- Encoding categorical variables improved training performance.
- Data preprocessing improved overall classification accuracy.

The analysis provided the following insights:

- Credit score strongly affects loan approval status.
- Applicants with higher income have a better chance of loan approval.
- Employment status and loan amount also influence approval decisions.
- Random Forest and Decision Tree performed significantly better than Logistic Regression.

The project demonstrated the practical application of machine learning and business analytics in the banking and finance sector for loan approval prediction.

Conclusion

In this project, machine learning techniques were successfully used to predict loan approval status using the Loan Prediction Dataset.

The project involved data preprocessing, feature engineering, model training, and comparative analysis of classification models. Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) were implemented and evaluated based on prediction accuracy.

Among all the implemented models, Random Forest and Decision Tree achieved the best performance with the highest accuracy and lowest prediction errors.

This project demonstrated how business analytics and machine learning can help financial institutions automate loan approval analysis and improve decision-making. In the future, the project can be improved further by implementing advanced deep learning algorithms and deploying the system as a web application.

Reference

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